

```
=====
Q01_Declare_Voltage_Current_Resistance.cpp
=====
```

```
// Assignment 1 - Question 1
```

```
// Name: Zawar Shah
```

```
// Registration No: BF25NWELE0677
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    double voltage, current, resistance; // Declare electrical quantities as double for decimal
```

```
    cout << "Variables voltage, current, and resistance are declared as double." << endl;
```

```
    return 0;
```

```
}
```

```
=====
Q02_Display_Welcome_Message.cpp
=====
```

```
// Assignment 1 - Question 2
```

```
// Name: Zawar Shah
```

```
// Registration No: BF25NWELE0677
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    cout << "Welcome to Object Oriented Programming in C++!" << endl; // Display the required
```

```
    return 0;
```

```
}
```

```
=====
Q03_Read_Age.cpp
=====
```

```
// Assignment 1 - Question 3
```

```
// Name: Zawar Shah
```

```
// Registration No: BF25NWELE0677
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int age;
```

```
    cout << "Enter your age: ";
```

```
    cin >> age; // Read integer value entered by the user
```

```
    cout << "Age entered: " << age << endl;
```

```
    return 0;
```

```
}
```

```

=====
Q04_Overheating_Detected.cpp
=====
// Assignment 1 - Question 4
// Name: Zawar Shah
// Registration No: BF25NWELE0677

#include <iostream>
using namespace std;

int main() {
    double temperature;

    cout << "Enter temperature: ";
    cin >> temperature;

    if (temperature > 100) { // Print warning only when temperature exceeds 100
        cout << "Overheating detected!" << endl;
    }

    return 0;
}

```

```

=====
Q05_Area_Of_Circle.cpp
=====
// Assignment 1 - Question 5
// Name: Zawar Shah
// Registration No: BF25NWELE0677

#include <iostream>
using namespace std;

int main() {
    double r, area;

    cout << "Enter radius of the circle: ";
    cin >> r;

    area = 3.14159 * r * r; // Formula for area of a circle: pi * r^2

    cout << "Area of circle: " << area << endl;
    return 0;
}

```

```

=====
Q06_Print_X_Y_Z_With_Commas.cpp
=====

```

```
// Assignment 1 - Question 6
// Name: Zawar Shah
// Registration No: BF25NWELE0677
```

```
#include <iostream>
using namespace std;
```

```
int main() {
    int x, y, z;

    cout << "Enter values of x, y, and z: ";
    cin >> x >> y >> z; // Read all three values in a single input statement

    cout << x << ", " << y << ", " << z << endl; // Print values separated by commas
    return 0;
}
```

```
=====
Q07_Sum_Of_Voltage_And_Current.cpp
=====
```

```
// Assignment 1 - Question 7
// Name: Zawar Shah
// Registration No: BF25NWELE0677
```

```
#include <iostream>
using namespace std;
```

```
int main() {
    double voltage, current, total;

    cout << "Enter voltage: ";
    cin >> voltage;

    cout << "Enter current: ";
    cin >> current;

    total = voltage + current; // Store the sum in variable total

    cout << "Total = " << total << endl;
    return 0;
}
```

```
=====
Q08_Division_By_Zero_Error.cpp
=====
```

```
// Assignment 1 - Question 8
// Name: Zawar Shah
// Registration No: BF25NWELE0677
```

```
#include <iostream>
using namespace std;
```

```

int main() {
    double denominator;

    cout << "Enter denominator: ";
    cin >> denominator;

    if (denominator == 0) { // Error message appears only when denominator is zero
        cout << "Error: Division by zero" << endl;
    }

    return 0;
}

```

```

=====
Q09_Increment_Count_And_Print.cpp
=====

```

```

// Assignment 1 - Question 9
// Name: Zawar Shah
// Registration No: BF25NWELE0677

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    int count;

    cout << "Enter count: ";
    cin >> count;

    count++; // Increase count by 1 before printing

    cout << "New value of count: " << count << endl;
    return 0;
}

```

```

=====
Q10_Read_Three_Floating_Point_Numbers.cpp
=====

```

```

// Assignment 1 - Question 10
// Name: Zawar Shah
// Registration No: BF25NWELE0677

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    float a, b, c;

    cout << "Enter three floating-point numbers: ";
    cin >> a >> b >> c; // Read three decimal numbers from the keyboard
}

```

```

    cout << "a = " << a << ", b = " << b << ", c = " << c << endl;
    return 0;
}

```

```

=====
Q11_Total_Resistance_In_Series.cpp
=====

```

```

// Assignment 1 - Question 11
// Name: Zawar Shah
// Registration No: BF25NWELE0677

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    double r1, r2, r3, totalResistance;

    cout << "Enter resistance R1: ";
    cin >> r1;
    cout << "Enter resistance R2: ";
    cin >> r2;
    cout << "Enter resistance R3: ";
    cin >> r3;

    totalResistance = r1 + r2 + r3; // In series, resistances are added directly

    cout << "Total resistance in series = " << totalResistance << " ohms" << endl;
    return 0;
}

```

```

=====
Q12_Total_Resistance_In_Parallel.cpp
=====

```

```

// Assignment 1 - Question 12
// Name: Zawar Shah
// Registration No: BF25NWELE0677

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    double r1, r2, r3, totalResistance;

    cout << "Enter resistance R1: ";
    cin >> r1;
    cout << "Enter resistance R2: ";
    cin >> r2;
    cout << "Enter resistance R3: ";
    cin >> r3;
}

```

```

        totalResistance = 1 / ((1 / r1) + (1 / r2) + (1 / r3)); // Equivalent resistance formula

        cout << "Total resistance in parallel = " << totalResistance << " ohms" << endl;
        return 0;
}

```

```

=====
Q13_Voltage_Division_Rule.cpp
=====

```

```

// Assignment 1 - Question 13
// Name: Zawar Shah
// Registration No: BF25NWELE0677

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    double r1 = 10.0;
    double r2 = 20.0;
    double voltage = 12.0;
    double voltageAcrossR2;

    voltageAcrossR2 = voltage * (r2 / (r1 + r2)); // Voltage across R2 in a series circuit

    cout << "Selected values: R1 = " << r1 << " ohms, R2 = " << r2 << " ohms, Voltage = " <<
    cout << "Voltage across R2 = " << voltageAcrossR2 << " volts" << endl;
    return 0;
}

```

```

=====
Q14_Current_Division_Rule.cpp
=====

```

```

// Assignment 1 - Question 14
// Name: Zawar Shah
// Registration No: BF25NWELE0677

```

```

#include <iostream>
using namespace std;

```

```

int main() {
    double r1 = 6.0;
    double r2 = 3.0;
    double voltage = 12.0;
    double totalCurrent, currentThroughR2;

    totalCurrent = voltage / ((r1 * r2) / (r1 + r2)); // Find total current using equivalent
    currentThroughR2 = totalCurrent * (r1 / (r1 + r2)); // Current division formula for the b

    cout << "Selected values: R1 = " << r1 << " ohms, R2 = " << r2 << " ohms, Voltage = " <<
    cout << "Current through R2 = " << currentThroughR2 << " amperes" << endl;
    return 0;
}

```

```
}
```

```
=====
Q15_Semester_GPA_Calculator.cpp
=====
```

```
// Assignment 1 - Question 15
// Name: Zawar Shah
// Registration No: BF25NWELE0677
```

```
#include <iostream>
#include <string>
using namespace std;
```

```
int main() {
    int numberOfSubjects;
    double creditHours, gradePoints;
    double totalQualityPoints = 0, totalCreditHours = 0, sgpa;
    string subjectName;

    cout << "Enter total number of subjects: ";
    cin >> numberOfSubjects;
    cin.ignore(); // Clear newline before using getline for subject names

    for (int i = 1; i <= numberOfSubjects; i++) {
        cout << "Enter subject " << i << " name: ";
        getline(cin, subjectName);

        cout << "Enter credit hours for " << subjectName << ": ";
        cin >> creditHours;

        cout << "Enter grade points for " << subjectName << ": ";
        cin >> gradePoints;

        totalQualityPoints += creditHours * gradePoints; // Quality points = credit hours multiplied by grade points
        totalCreditHours += creditHours;
        cin.ignore(); // Clear newline before reading the next subject name
    }

    sgpa = totalQualityPoints / totalCreditHours; // SGPA = total quality points / total credit hours

    cout << "Semester GPA = " << sgpa << endl;
    return 0;
}
```